Lecture 4: Parametric Classification

Spring 2026

10 ECTS Credits

Department of Information and Communication Technology

Faculty of Engineering and Science

UiA

Recap

- Classification Rules
  - Nearest Centroid
  - Nearest-Neighbor
  - Discrete Histogram
- Classification Error Rates
- Consistency
  - Weak consistency
  - Strong consistency
- Consistency in Terms of the Expected Error
- No-Free-Lunch Theorems

Parameter Estimation

What is a parameter?

- A value that defines a probability distribution.
- Examples:
  - Coin Flip: $p = P(\text{Heads})$
  - Gaussian Distribution: μ (mean), σ (standard deviation)
  - Linear regression: $Y = mX + c$

    m (slope) and c (intercept) explains the relationship between X and Y.
- Problem: Given observed data, find parameter values that best explain the data.
- Let the parameters be represented by θ.

Parameter Estimation

- Given: Data $X_1, X_2, \ldots, X_n$
- Goal: Estimate θ from data assuming data comes from a distribution with parameter θ.
- A coin is flipped 3 times resulting in H, H, T. What is the probability of getting 2 heads in 3 flips?

$$P(\text{2 Heads in 3 Flips} \mid p = 0.5) = \binom{3}{2} p^2 (1-p)^1$$

$$= \frac{3!}{2! \times 1!} \times 0.5^2 \times 0.5 = 0.375$$

- But what if we want to know the value of p that explains observing 2 heads in 3 flips?

Likelihood Function

- Probability $P(\text{Data} \mid \text{Parameter})$: Keep θ fixed
- Likelihood $L(\text{Parameter} \mid \text{Data})$: Keep data fixed to calculate θ.
- $p = \frac{1}{2}$:

$$L(0.5) = \binom{3}{2} p^2 (1-p)^1 = 3 \times 0.5^2 \times 0.5 = 0.375$$

If $p = \frac{2}{3}$:

$$L(0.6\bar{6}) = \binom{3}{2} p^2 (1-p)^1 = 3 \times 0.6\bar{6}^2 \times 0.6\bar{6} \approx 0.444$$

$p = 0.6\bar{6}$ has a higher likelihood implying $p = 2/3$ better explains how you got 2 heads and 1 tail on flipping the coin 3 times.

Maximum Likelihood Estimation

- Given independent observations $x_1, x_2, \dots, x_n$, the likelihood is given by:

$$L(\theta) = \prod_{i=1}^{n} P(\, x_i \mid \theta \,)$$

- For continuous distributions, we replace $P(\, x_i \mid \theta \,)$ with probability density $p(\, x_i \mid \theta \,)$.
- Taking Log, we get the log-likelihood:

$$\ln\big(L(\theta)\big) = \sum_{i=1}^{n} \ln\big(P(\, x_i \mid \theta \,)\big)$$

- Solving $\frac{d}{d\theta}\Big(\ln\big(L(\theta)\big)\Big) = 0$ gives θ that maximizes $L(\theta)$

MLE for Gaussian Distribution

Five students have heights $[165, 170, 168, 172, 169]$ in cm. Assuming heights follow a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$. Find the ML estimates $\hat{\mu}$ and $\hat{\sigma}^2$. What distribution best explains the data?

- Step 1: Calculate Likelihood

$$p(x_i \mid \mu, \sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

For all observations,

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{(2\pi\sigma^2)}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

MLE for Gaussian Distribution

- Step 2: Calculate Log-likelihood

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{(2\pi\sigma^2)}} \cdot \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$\ln\big(L(\mu, \sigma^2)\big) = \ln\left[\prod_{i=1}^{n} \frac{1}{\sqrt{(2\pi\sigma^2)}} \cdot \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)\right]$$

$$\ln\big(L(\mu, \sigma^2)\big) = \sum_{i=1}^{n} \ln\left[\frac{1}{\sqrt{(2\pi\sigma^2)}} \cdot \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)\right]$$

MLE for Gaussian Distribution

$$\ln\left(L(\mu,\sigma^2)\right) = \sum_{i=1}^{n}\left[\ln\left(\frac{1}{\sqrt{(2\pi\sigma^2)}}\right) + \ln\left(\exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)\right)\right]$$

$$\ln\left(L(\mu,\sigma^2)\right) = \sum_{i=1}^{n}\left[\ln\left(\frac{1}{\sqrt{(2\pi\sigma^2)}}\right) - \frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

$$\ln\left(\frac{1}{\sqrt{(2\pi\sigma^2)}}\right) = -\ln\left(\sqrt{(2\pi\sigma^2)}\right) = -\frac{1}{2}\ln(2\pi\sigma^2) = -\frac{1}{2}[\ln(2\pi) + \ln(\sigma^2)]$$

$$\ln\left(L(\mu,\sigma^2)\right) = \sum_{i=1}^{n}\left[-\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln(\sigma^2) - \frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

MLE for Gaussian Distribution

$$\ln\big(L(\mu,\sigma^2)\big) = \sum_{i=1}^{n}\left[-\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln(\sigma^2) - \frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

$$\ln\big(L(\mu,\sigma^2)\big) = -\sum_{i=1}^{n}\left[\frac{1}{2}\ln(2\pi)\right] - \sum_{i=1}^{n}\left[\frac{1}{2}\ln(\sigma^2)\right] - \sum_{i=1}^{n}\left[\frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

$$\ln\big(L(\mu,\sigma^2)\big) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}[(x_i-\mu)^2]$$

MLE for Gaussian Distribution

- Step 3: Maximize by taking derivatives and setting to 0.

  For μ,

$$\frac{\partial}{\partial\mu}\left[\ln\left(L(\mu,\sigma^2)\right)\right] = \frac{\partial}{\partial\mu}\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}[(x_i-\mu)^2]\right]$$

$$\frac{\partial}{\partial\mu}\left[\ln\left(L(\mu,\sigma^2)\right)\right] = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[\frac{\partial}{\partial\mu}[(x_i-\mu)^2]\right]$$

$$\frac{\partial}{\partial\mu}\left[\ln\left(L(\mu,\sigma^2)\right)\right] = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[2(x_i-\mu)\cdot\frac{\partial}{\partial\mu}(x_i-\mu)\right]$$

$$\frac{\partial}{\partial\mu}\left[\ln\left(L(\mu,\sigma^2)\right)\right] = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$$

MLE for Gaussian Distribution

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$

$$\sum_{i=1}^{n} x_i - n\mu = 0$$

$$\sum_{i=1}^{n} x_i = n\mu$$

$\hat{\mu}_{ML} = \frac{1}{n}\sum_{i=1}^{n} x_i$, (sample mean!)

MLE for Gaussian Distribution

Similarly for $\widehat{\sigma^2}_{ML}$, take partial derivative with respect to σ^2 and equate to 0.

$$\frac{\partial}{\partial\sigma^2}\left[\ln\left(L(\mu,\sigma^2)\right)\right] = \frac{\partial}{\partial\sigma^2}\left[-\frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}[(x_i-\mu)^2]\right]$$

$$\frac{\partial}{\partial\sigma^2}\left[\ln\left(L(\mu,\sigma^2)\right)\right] = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}[(x_i-\mu)^2]$$

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}[(x_i-\mu)^2] = 0$$

$\widehat{\sigma^2}_{ML} = \frac{1}{n}\sum_{i=1}^{n}[(x_i-\mu)^2]$, (sample variance!)

MLE for Gaussian Distribution

- Step 4: Calculate using data

$$\hat{\mu}_{ML} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{(165\ +\ 170\ +\ 168\ +\ 172\ +\ 169)}{5} = 168.8\ \mathrm{cm}$$

$$\widehat{\sigma^2}_{ML} = \frac{1}{n}\sum_{i=1}^{n}[(x_i - \mu)^2] = \frac{[(165 - 168.8)^2 + \ldots + (169 - 168.8)^2]}{5} = 5.36\ \mathrm{cm}^2$$

- Result: The distribution is estimated to be $\mathcal{N}(168.8, 5.36)$

MLE for Classification

- We used MLE to estimate parameters for distribution:
  - Coin → estimate p,
  - Gaussian → estimate μ and σ^2
- For classification, we estimate parameters for per class distribution:
  - Class 0 → estimate θ_0
  - Class 1 → estimate θ_1
- Then, we compare these two per class distribution weighted by priors to assign class 1 if:
$$p(\boldsymbol{x} \mid Y = 1; \theta_1)P(Y = 1) > p(\boldsymbol{x} \mid Y = 0; \theta_0)P(Y = 0)$$
- Why? According to Bayes' Theorem: $P(Y = 1 \mid \boldsymbol{x}) \propto p(\boldsymbol{x} \mid Y = 1)P(Y = 1)$.

Parametric Plug-in Rules

- "Plug-in": estimate parameters from data and plug them into Bayes classifier to create a practical classification rule.
- Step 1: Estimate per class distribution parameters using MLE
  - Use class 0 data to estimate θ_0
  - Use class 1 data to estimate θ_1
  - Estimate priors.
- Step 2: Plug estimates into Bayes rule

$$P(Y = c \mid \boldsymbol{x}) \propto p(\boldsymbol{x} \mid Y = c; \theta_c) P(Y = c)$$

- Step 3: Choose class with higher probability.

Parametric Plug-in Rules

- Bayes Classifier (1):

$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & D^*(\boldsymbol{x}) > k^* \\ 0, & \text{otherwise} \end{cases}$$

where the **optimal discriminant** $D^*: R^d \rightarrow R$ is given by (2):

$$D^*(\boldsymbol{x}) = \ln \frac{p(\boldsymbol{x} \mid Y = 1)}{p(\boldsymbol{x} \mid Y = 0)}$$

for $x \in R^d$, with **optimal threshold** k^* given by

$$k^* = \ln \frac{P(Y=0)}{P(Y=1)}$$

Parametric Plug-in Rules

- Assume knowledge about $P_{\boldsymbol{X},Y}$ is coded into a family of probability density functions $\{p(\boldsymbol{x} \mid \boldsymbol{\theta}) \mid \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^m\}$:
  - $p(\boldsymbol{x} \mid \boldsymbol{\theta}_0^*)$, where $\boldsymbol{\theta}_0^*$ is the *true* parameter value for class 0, and
  - $p(\boldsymbol{x} \mid \boldsymbol{\theta}_1^*)$, where $\boldsymbol{\theta}_1^*$ is the *true* parameter value for class 1.
- We assume, $\boldsymbol{\theta}_0^*$ and $\boldsymbol{\theta}_1^*$ could be estimated using $\boldsymbol{\theta}_{0,n}$ and $\boldsymbol{\theta}_{1,n}$ from sample data $S_n = \{(\boldsymbol{X}_1, Y_1), \ldots (\boldsymbol{X}_n, Y_n)\}$.
- The **parametric plug-in sample-based discriminant** can be obtained by plugging in these estimators in the optimal discriminant equation.

Parametric Plug-in Rules

- From (2), optimal discriminant becomes:

$$D^*(\boldsymbol{x}) = \ln \frac{p(\boldsymbol{x} \mid \boldsymbol{\theta}_{1,n})}{p(\boldsymbol{x} \mid \boldsymbol{\theta}_{0,n})}$$

- And from (1), the parametric plug-in classifier becomes:

$$\psi_n(\boldsymbol{x}) = \begin{cases} 1, & D_n(\boldsymbol{x}) > k_n \\ 0, & \text{otherwise.} \end{cases}$$

- For obtaining the threshold k_n,
  1. If $c_0 = P(Y = 0)$ and $c_1 = P(Y = 1)$ are known, use $k^* = \ln {}^{c_0}/_{c_1}$
  2. If not known, but training sample size is moderate to large and sampling is random, use the estimate

$$k_n = \ln \frac{N_0/n}{N_1/n} = \ln \frac{N_0}{N_1}$$

Parametric Plug-in Rules

3. If c_0 and c_1 are unknown and sample size is small, or sampling is not random, use minimax approach to obtain k_n.
4. Vary k_n and search for the best value using error estimation methods.
5. In cases where sample size is small due to cost of labelling data, but unlabelled data are abundant:
   a. Train a classifier ψ on the available labeled data (S_n)
   b. Feed unlabeled data through the classifier to get proportions $R_{0,m}$ and $R_{1,m}$
   c. Calculate k_n using the relationship $k_n = \ln {}^{\hat{c}_0}/_{\hat{c}_1}$, where $\hat{c}_0$ and $\hat{c}_1$ are derived from $R_{0,m}$ and $R_{1,m}$.

Parametric Plug-in Rules

- Another way is assuming a posterior probability function $\eta(\boldsymbol{x} \mid \boldsymbol{\theta}^*)$ as a member of parametric family $\{\eta(\boldsymbol{x} \mid \boldsymbol{\theta}^*) \mid \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^m\}$:

$$\psi_n(\boldsymbol{x}) = \begin{cases} 1, & \eta(\boldsymbol{x} \mid \boldsymbol{\theta}^*) > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

- *Advantage*: Avoids dealing with multiple parameters $\boldsymbol{\theta}_0^*$ and $\boldsymbol{\theta}_1^*$, and choice of discriminant threshold k_n.
- *Disadvantage*: It is easier to model class-conditional densities than posterior probability function directly.

Gaussian Discriminant Analysis

- Consider the class-conditional feature distribution $P(\boldsymbol{x} \mid Y = y)$ parametrized by mean $\boldsymbol{\mu}$ and covariance matrix Σ:

$$p(\boldsymbol{x} \mid \boldsymbol{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x}-\boldsymbol{\mu})\right]$$

- Here the distribution is multivariate Gaussian, and the case is referred to as Gaussian Discriminant Analysis.
- From the above class-conditional feature distribution, the optimal discriminant for a Bayes classifier is given by:

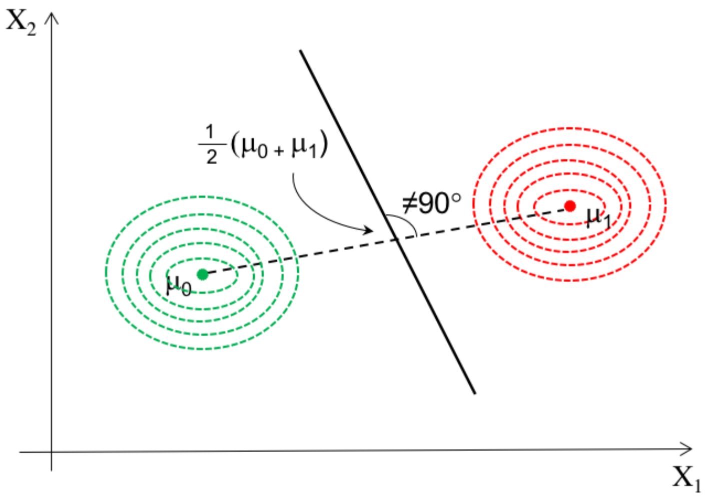
$$D^*(\boldsymbol{x}) = \underbrace{\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_0)^T \Sigma_0^{-1} (\boldsymbol{x}-\boldsymbol{\mu}_0)}_{\text{Squared Mahalanobis distance from } \boldsymbol{x} \text{ to class 0 center}} - \underbrace{\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_1)^T \Sigma_1^{-1} (\boldsymbol{x}-\boldsymbol{\mu}_1)}_{\text{Squared Mahalanobis distance from } \boldsymbol{x} \text{ to class 1 center}} + \underbrace{\frac{1}{2}\ln\frac{|\Sigma_0|}{|\Sigma_1|}}_{\text{Penalty for different covariance}}$$

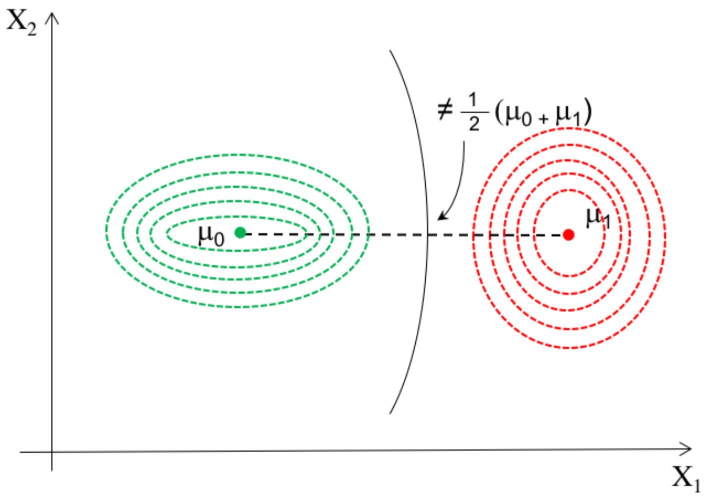
Gaussian Discriminant Analysis

- Plugging in the estimators $(\hat{\boldsymbol{\mu}}, \hat{\Sigma})$ for the true parameters $(\boldsymbol{\mu}, \Sigma)$, we have (3):

$$D_n(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{x} - \hat{\boldsymbol{\mu}}_0)^T \hat{\Sigma}_0^{-1}(\boldsymbol{x} - \hat{\boldsymbol{\mu}}_0) - \frac{1}{2}(\boldsymbol{x} - \hat{\boldsymbol{\mu}}_1)^T \hat{\Sigma}_1^{-1}(\boldsymbol{x} - \hat{\boldsymbol{\mu}}_1) + \frac{1}{2}\ln\frac{|\hat{\Sigma}_0|}{|\hat{\Sigma}_1|}$$

- If $\Sigma_0 = \Sigma_1 = \Sigma$, we have **Linear Discriminant Analysis** and if $\Sigma_0 \neq \Sigma_1$, we have **Quadratic Discriminant Analysis**.





Linear Discriminant Analysis

$$D_n(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_0)^T \widehat{\Sigma}_0^{-1}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_0) - \frac{1}{2}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_1)^T \widehat{\Sigma}_1^{-1}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_1) + \frac{1}{2}\ln\frac{|\widehat{\Sigma}_0|}{|\widehat{\Sigma}_1|}$$

- The MLE of means vectors are given by sample means:

  $$\widehat{\boldsymbol{\mu}}_0 = \frac{1}{N_0}\sum_{i=1}^{n} \boldsymbol{X}_i I_{Y_i=0} \text{ and } \widehat{\boldsymbol{\mu}}_1 = \frac{1}{N_1}\sum_{i=1}^{n} \boldsymbol{X}_i I_{Y_i=1}$$

  where N_0 and N_1 are the class-specific sample sizes
- Under the assumption $\Sigma_0 = \Sigma_1 = \Sigma$, the MLE of Σ becomes:

$$\widehat{\Sigma}^{ML} = \frac{N_0}{n}\widehat{\Sigma}_0^{ML} + \frac{N_1}{n}\widehat{\Sigma}_1^{ML}$$

$$\widehat{\Sigma}_0^{ML} = \frac{1}{N_0}\sum_{i=1}^{n}\left[(\boldsymbol{X}_i - \hat{\mu}_0)(\boldsymbol{X}_i - \hat{\mu}_0)^T I_{Y_i=0}\right], \qquad \widehat{\Sigma}_1^{ML} = \frac{1}{N_1}\sum_{i=1}^{n}\left[(\boldsymbol{X}_i - \hat{\mu}_1)(\boldsymbol{X}_i - \hat{\mu}_1)^T I_{Y_i=1}\right]$$

$$\widehat{\Sigma}^{ML} = \frac{1}{n}\sum_{i=1}^{n}\left[(\boldsymbol{X}_i - \hat{\mu}_0)(\boldsymbol{X}_i - \hat{\mu}_0)^T I_{Y_i=0} + (\boldsymbol{X}_i - \hat{\mu}_1)(\boldsymbol{X}_i - \hat{\mu}_1)^T I_{Y_i=1}\right]$$

Linear Discriminant Analysis

- This estimator is biased. An unbiased estimator is given by (ref):

$$\hat{\Sigma} = \frac{n}{n-2}\hat{\Sigma}^{ML} = \frac{(N_0 - 1)\hat{\Sigma}_0 + (N_1 - 1)\hat{\Sigma}_1}{n-2}$$

  where $\hat{\Sigma}_0$ and $\hat{\Sigma}_1$ are the sample covariance matrices
- The estimator is known as **pooled sample covariance matrix.**
- The LDA discriminant can be obtained by substituting the above in (3):

$$D_{L,n}(\boldsymbol{x}) = (\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\Sigma}^{-1}\left(\boldsymbol{x} - \frac{\hat{\boldsymbol{\mu}}_0 + \hat{\boldsymbol{\mu}}_1}{2}\right)$$

- This discriminant is also known as **Anderson's W statistic**.

Linear Discriminant Analysis

- The LDA classifier produces a *hyperplane decision boundary*, determined by the equation $\boldsymbol{a}_n^T \boldsymbol{x} + b_n = k_n$, where (4):

$$\boldsymbol{a}_n = \hat{\Sigma}^{-1}(\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0),$$

$$b_n = (\hat{\boldsymbol{\mu}}_0 - \hat{\boldsymbol{\mu}}_1)\hat{\Sigma}^{-1}\left(\frac{\hat{\boldsymbol{\mu}}_0 + \hat{\boldsymbol{\mu}}_1}{2}\right)$$

- If $P(Y = 0)$ and $P(Y = 1)$ are unknown and sample size is small, or sampling is non-random, a common choice for LDA threshold is $k_n = 0$.
- $k_n = 0$ implies the decision hyperplane passes through the midpoint between the sample means:

$$\hat{\boldsymbol{x}}_m = \frac{\hat{\boldsymbol{\mu}}_0 + \hat{\boldsymbol{\mu}}_1}{2}$$

Eigenvectors (Recall)

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

ρ is the correlation, σ_1 and σ_2 are the standard deviation.

Any symmetric matrix Σ $(d \times d)$ can be decomposed as:

$$\Sigma = V\Lambda V^T$$

V is the $d \times d$ matrix where each column is an eigenvector, Λ is the $d \times d$ diagonal matrix of eigenvalues λ_i

$$\Sigma^{-1} = V\Lambda^{-1}V^T$$

Λ^{-1} is the $d \times d$ diagonal matrix of ${}^{1}/_{\lambda_i}$.

Linear Discriminant Analysis

- Number of parameters involved in estimating pooled sample covariance matrix: $d + {}^{d(d-1)}/_2 = {}^{d(d+1)}/_2$
- As dimensionality d approaches number of training samples n:
  - large eigenvalues (λ) of Σ: $\lambda_{estimated} > \lambda_{\text{true}}$
  - smaller eigenvalues (λ) of Σ: $\lambda_{estimated} < \lambda_{\text{true}}$.
- Smaller eigenvalues being underestimated leads to $\det|\hat{\Sigma}| \approx 0$ making inverse computation unstable.
- To avoid this, we use some variants of LDA in small-sample cases.

Linear Discriminant Analysis - Variants

- **Diagonal LDA**: The estimator of Σ will be constrained to be a diagonal matrix $\hat{\Sigma}_D$ such that:

$$\left(\hat{\Sigma}_D\right)_{jj} = \left(\hat{\Sigma}\right)_{jj}, \text{ with } \left(\hat{\Sigma}_D\right)_{jk} = 0 \text{ for } j \neq k.$$

- **Nearest-Mean Classifier (NMC)**: The estimator of Σ will be constrained to be a diagonal matrix $\hat{\Sigma}_M$ with equal diagonal elements commonly:

$$\left(\hat{\Sigma}_M\right)_{jj} = \hat{\sigma}^2 = \frac{1}{d}\sum_{k=1}^{d}\left(\hat{\Sigma}\right)_{kk} \text{ with } \left(\hat{\Sigma}_M\right)_{jk} = 0 \text{ for } j \neq k.$$

  From (4), $\boldsymbol{a}_n$ and b_n contains the term $\frac{1}{\sigma^2}$. With $k_n = 0$, the term drops out and the classifier depends only on sample means.

- **Covariance Plug-In**: If Σ is assumed to be known or could be guessed, it can be used instead of $\hat{\Sigma}$. Only sample means need to be estimated then.

Quadratic Discriminant Analysis

- Under the assumption $\Sigma_0 \neq \Sigma_1$:

$$D_n(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_0)^T \hat{\Sigma}_0^{-1}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_0) - \frac{1}{2}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_1)^T \hat{\Sigma}_1^{-1}(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_1) + \frac{1}{2}\ln\frac{|\hat{\Sigma}_0|}{|\hat{\Sigma}_1|}$$

- The QDA classifier produces a hyperplane decision boundary, determined by the equation $\boldsymbol{x}^T \boldsymbol{A}_n \boldsymbol{x} + \boldsymbol{b}_n^T \boldsymbol{x} + c_n = k_n$, where:

$$\boldsymbol{A}_n = -\frac{1}{2}\left(\hat{\Sigma}_1^{-1} - \hat{\Sigma}_0^{-1}\right),$$

$$\boldsymbol{b}_n = \hat{\Sigma}_1^{-1}\widehat{\boldsymbol{\mu}}_1 - \hat{\Sigma}_0^{-1}\widehat{\boldsymbol{\mu}}_{\boldsymbol{0}}$$

$$c_n = -\frac{1}{2}\left(\widehat{\boldsymbol{\mu}}_1^T \hat{\Sigma}_1^{-1}\widehat{\boldsymbol{\mu}}_1 - \widehat{\boldsymbol{\mu}}_0^T \hat{\Sigma}_0^{-1}\widehat{\boldsymbol{\mu}}_{\boldsymbol{0}}\right) - \frac{1}{2}\ln\frac{|\hat{\Sigma}_1|}{|\hat{\Sigma}_0|}$$

Logistic Classification

- Logit transformation is defined for $0 < p < 1$:

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

which maps the bounded $[0,1]$ interval into the real line $(-\infty, +\infty)$.

- In logit space, the posterior probability function is linear, with parameters $(\boldsymbol{a}, b)$:

$$\text{logit}\big(\eta(\boldsymbol{x} \mid \boldsymbol{a}, b)\big) = \ln\left(\frac{\eta(\boldsymbol{x} \mid \boldsymbol{a}, b)}{1-\eta(\boldsymbol{x} \mid \boldsymbol{a}, b)}\right) = \boldsymbol{a}^T\boldsymbol{x} + b.$$

Taking exponential on both sides:

$$\frac{\eta(\boldsymbol{x} \mid \boldsymbol{a}, b)}{1-\eta(\boldsymbol{x} \mid \boldsymbol{a}, b)} = e^{\boldsymbol{a}^T\boldsymbol{x}+b}$$

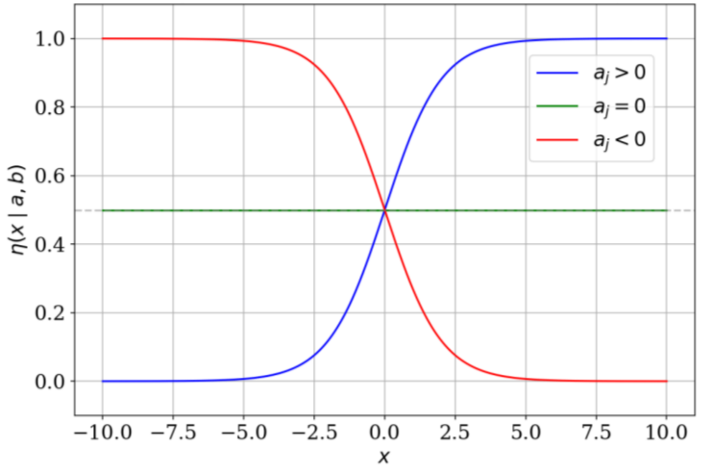
$$\eta(\boldsymbol{x} \mid \boldsymbol{a}, b) = e^{\boldsymbol{a}^T\boldsymbol{x}+b}\big(1-\eta(\boldsymbol{x} \mid \boldsymbol{a}, b)\big) \Rightarrow \eta(\boldsymbol{x} \mid \boldsymbol{a}, b)\left(1 + e^{\boldsymbol{a}^T\boldsymbol{x}+b}\right) = e^{\boldsymbol{a}^T\boldsymbol{x}+b}$$

$\eta(\boldsymbol{x} \mid \boldsymbol{a}, b) = \frac{e^{\boldsymbol{a}^T\boldsymbol{x}+b}}{1+e^{\boldsymbol{a}^T\boldsymbol{x}+b}} = \frac{1}{1+e^{-(\boldsymbol{a}^T\boldsymbol{x}+b)}}$, obtained by multiplying $e^{-(\boldsymbol{a}^T\boldsymbol{x}+b)}$ in numerator and denominator

Logistic Classification

$$\eta(\,\boldsymbol{x} \mid \boldsymbol{a}, b\,) = \frac{1}{1 + e^{-(\boldsymbol{a}^T\boldsymbol{x}+b)}}$$

- The function is called **logistic curve** with parameters $(\boldsymbol{a}, b)$, which is
  - strictly increasing for a>0
  - strictly decreasing for a<0
  - constant for a=0
- The estimates $\boldsymbol{a}_\mathrm{n}$ and b_n are obtained by maximizing the conditional log-likelihood $L(\,\boldsymbol{a}, b \mid S_n\,)$ under parametric assumption $\eta(\,\boldsymbol{x} \mid \boldsymbol{a}, b\,)$.



Logistic Classification

$$L(\boldsymbol{a}, b \mid S_n) = \ln\left(\prod_{i=1}^{n} P(Y = y_i \mid \boldsymbol{X} = \boldsymbol{x})\right) = \sum_{i=1}^{n}\left[y_i(\boldsymbol{a}^T\boldsymbol{x} + b) - \ln\left(1 + e^{\boldsymbol{a}^T\boldsymbol{x}+b}\right)\right]$$

- Deriving the log-likelihood function with respect to $\boldsymbol{a}_n$ and b_n and equating to zero maximizes the conditional log-likelihood.
- Solving with iterative numerical models and plugging in the estimates, we get the logistic classifier:

$$\psi_n(\boldsymbol{x}) = \begin{cases} 1, & \dfrac{1}{1 + e^{-(\boldsymbol{a}_n^T\boldsymbol{x}+b_n)}} > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

Thank You